Electronic Engineering - Embedded Systems

Lab 1 Report

Numerical analysis of matching circuits

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1 Process description

1.1 Problem nr.1

The first thing I did was evaluate the characteristic impedances for the two different cases:

$$Z_{c1} = \sqrt{Z_0 \cdot Z_{L1}} = 70.10\ \Omega \tag{1}$$

$$Z_{c2} = \sqrt{Z_0 \cdot Z_{L2}} = 100.00\ \Omega \tag{2}$$

In the following steps, for the evaluation of points a) and b), it will not be possible to report all the values from the vectors, so I will just report all the formulas and the reasoning behind it.

Since I have to plot the reflection coefficient as the frequency varies, the first thing I did was creating a linspace that contains all the normalized frequencies from 0 to 2; this was necessary in order to calculate the electrical lenght of the quarter wave transformer line, the input impedances and the reflection coefficient at point A of the TL. With this done, I procedeed with the calculation of $S_{11_{dB}}$ which will be used in point b) of problem nr.1.

In the end I calculated the fractional bandwidth for both cases and got:
For case 1:

$$\frac{f_2 - f_1}{f_0} = [0.817, 1.183] \tag{3}$$

$$Fractional\ Bandwidth : 36.66\% \tag{4}$$

For case 2:

$$\frac{f_2 - f_1}{f_0} = [0.917, 1.086] \tag{5}$$

$$Fractional\ Bandwidth : 36.66\% \tag{6}$$

The resulting graphs are:

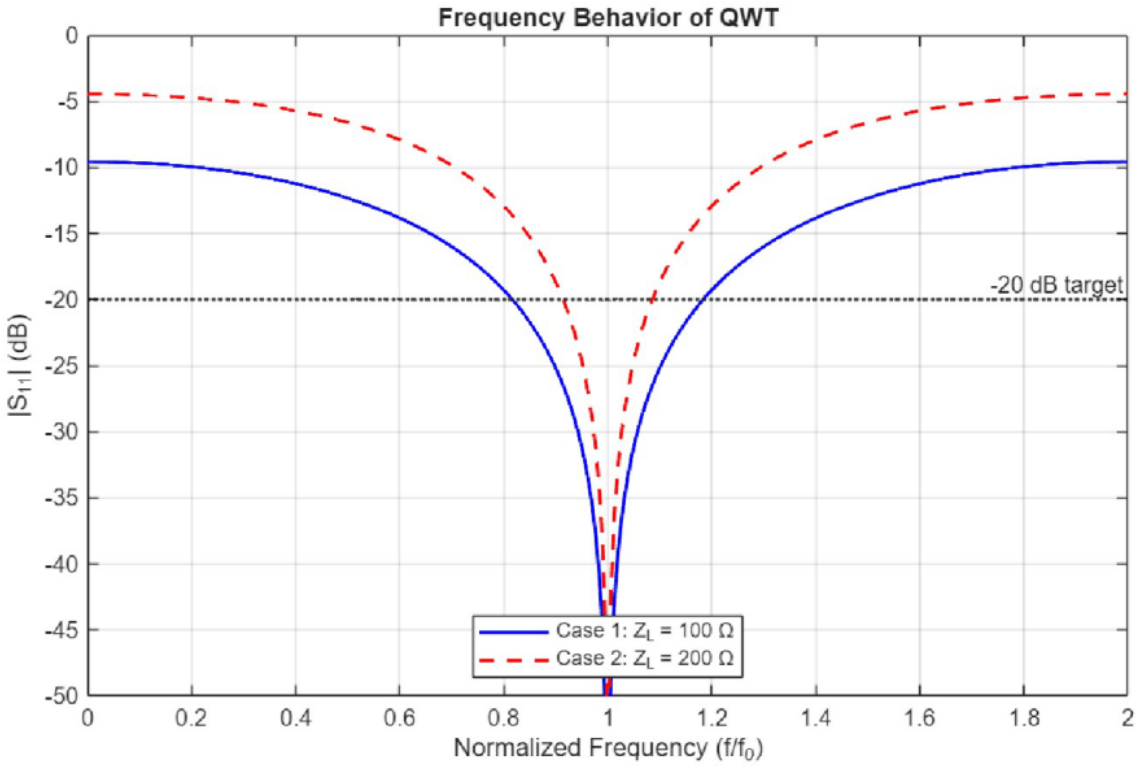

Figure 1: $|S_{11}|$ graphs for Problem nr.1

As expected for a Quarter-Wave Transformer $|S_{11}|$, which is the reflection coefficient, is null at the center band frequency; in this case the center band frequency is 1 because the frequency interval is normalized in respect to f_0. Another consideration that can be made is that, with the addition of a load impedance (Z_L) of 200 Ω, the result that we have is that the absolute value of the reflection coefficient is higher compared to the same transmission line with a load impedance of 100 Ω.

We can notice that Case 2 has a narrower frequency band where $|S_{11_{dB}}|$ is lower than $-20\ dB$.

1.2 Problem nr.2

Each line is a QWT, so I could use the exact same formula as Problem nr.1 to evaluate the electrical lenght of each transmission line. Then I procedeed with the calculation of the S matrix for both TLs and multiplied them in order to obtain the final S matrix of the entire system. The used formulas were:

$$\begin{aligned} A_i &= \cos\beta_i\ell_i & B_i &= jZ_i\sin\beta_i\ell_i \\ C_i &= jY_i\sin\beta_i\ell_i & D_i &= \cos\beta_i\ell_i \end{aligned} \tag{7}$$

Where:

$$1 \leq i \leq 2 \tag{8}$$

For the computation of the total matrix I used:

$$\begin{aligned} A_{\text{tot}} &= A_1 \cdot A_2 + B_1 \cdot C_2 \\ B_{\text{tot}} &= A_1 \cdot B_2 + B_1 \cdot D_2 \\ C_{\text{tot}} &= C_1 \cdot A_2 + D_1 \cdot C_2 \\ D_{\text{tot}} &= C_1 \cdot B_2 + D_1 \cdot D_2 \end{aligned} \tag{9}$$

From this, as for Problem nr.1 I proceeded with the calculation of $|S_{11_{dB}}|$ and then with the definition of points where the reflection coefficient is lower than $-20\ dB$.

Finally I calculated the fractional badwidth as:

$$\frac{f_2 - f_1}{f_0} = [0.628, 1.372] \tag{10}$$

$$Fractional\ Bandwidth : 74.35\% \tag{11}$$

By plotting the reflection coefficient as the frequency changes I got:

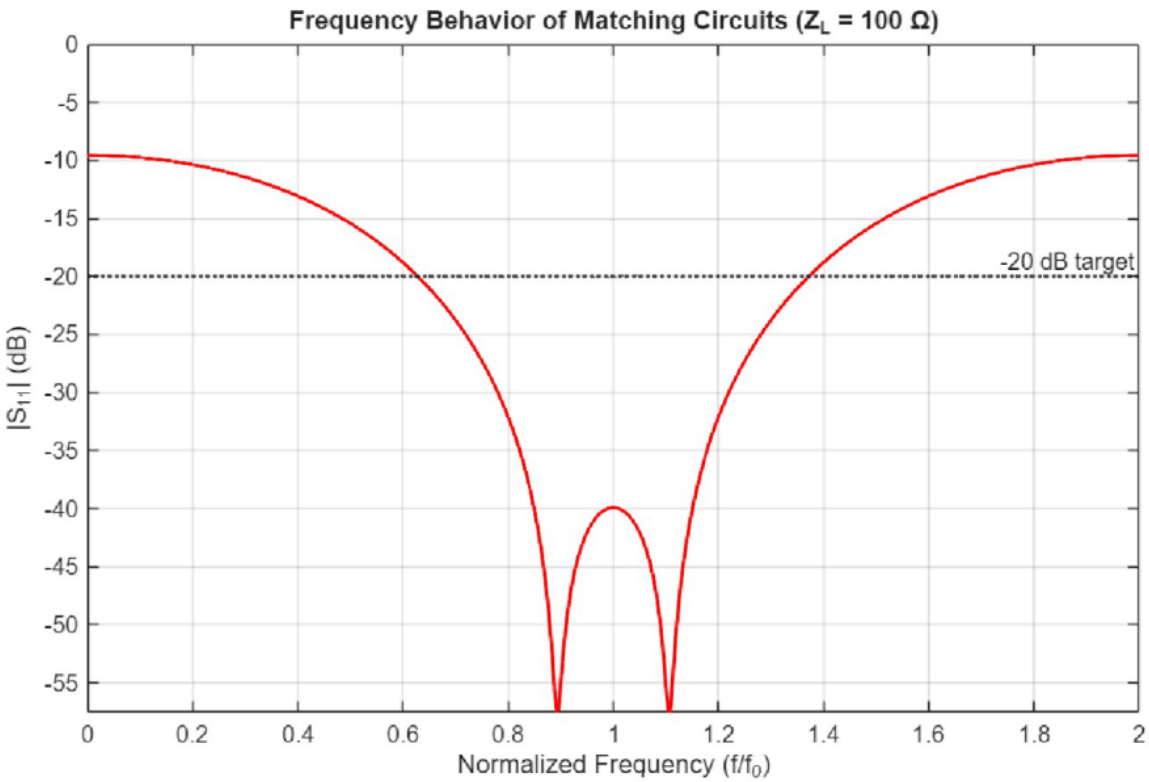

Figure 2: $|S_{11}|$ graphs for Problem nr.2

By reading the graph, we can notice that the maximum absolute value is exactly the

same as Case 1, Problem nr.1 (Blue line in Figure 1); but we can notice almost instantly the fact that the frequency band where $|S_{11_{dB}}|$ is lower than $-20\ dB$ is almost double the one for Case 1, Problem nr.1. In addition to that we can add that at the center band frequency, represented here by 1 since the frequency band has been normalized, there isn't anymore a null value for the reflection coeffiecient. Instead the cascade of transmission lines gave us two nulls left and right from the center band frequency f_0.

A Appendix

A.1 Problem nr.1 MATLAB code

```
clear; clc; close all;

%% Define Parameters
Z0 = 50;          % Characteristic impedance
ZL1 = 100;        % Load impedance for Case 1
ZL2 = 200;        % Load impedance for Case 2

% Calculate Zc for both cases
Zc1 = sqrt(Z0 * ZL1);
Zc2 = sqrt(Z0 * ZL2);

fprintf('--- Characteristic Impedances ---\n');
fprintf('Case 1 (ZL = 100 Ohms): Zc = %.2f Ohms\n', Zc1);
fprintf('Case 2 (ZL = 200 Ohms): Zc = %.2f Ohms\n\n', Zc2);

%% Normalized frequency range f_norm = f / f0
%  it has to be calculated for the range [0,2f0], that's why it goes from 0
%  to 2
f_norm = linspace(0, 2, 10000);

% Electrical length of a quarter-wave line
theta = (pi / 2) * f_norm;

%% Calculate S11
% Case 1
% Input impedance
```

```
Zin1 = Zc1 .* (ZL1 + 1j * Zc1 .* tan(theta)) ./ (Zc1 + 1j * ZL1 .* tan(theta));
% S11 and conversion to dB
S11_1 = (Zin1 - Z0) ./ (Zin1 + Z0);
S11_1_dB = 20 * log10(abs(S11_1));

% Case 2
Zin2 = Zc2 .* (ZL2 + 1j * Zc2 .* tan(theta)) ./ (Zc2 + 1j * ZL2 .* tan(theta));

S11_2 = (Zin2 - Z0) ./ (Zin2 + Z0);
S11_2_dB = 20 * log10(abs(S11_2));

%% Determine Fractional Bandwidth (S11 < -20 dB)
fprintf('--- Fractional Bandwidths ---\n');

% Case 1 Bandwidth
% we use the find function to fill idx_p2 of all the addresses of the
% values that satisfy the condition, then we pick the first and the last
% value.
idx1 = find(S11_1_dB < -20);
if ~isempty(idx1)
    f1_1 = f_norm(idx1(1));
    f2_1 = f_norm(idx1(end));
    BW1 = f2_1 - f1_1;
    fprintf('Case 1 (ZL = 100 Ohms):\n');
    fprintf('  Interval [f1, f2] / f0: [%.3f, %.3f]\n', f1_1, f2_1);
    fprintf('  Fractional Bandwidth: %.2f%%\n\n', BW1 * 100);
else
    fprintf('Case 1 never reaches |S11| < -20 dB.\n\n');
end

% Case 2 Bandwidth
idx2 = find(S11_2_dB < -20);
if ~isempty(idx2)
    f1_2 = f_norm(idx2(1));
```

```
    f2_2 = f_norm(idx2(end));
    BW2 = f2_2 - f1_2;
    fprintf('Case 2 (ZL = 200 Ohms):\n');
    fprintf('  Interval [f1, f2] / f0: [%.3f, %.3f]\n', f1_2,
       f2_2);
    fprintf('  Fractional Bandwidth: %.2f%%\n', BW2 * 100);
else
    fprintf('Case 2 never reaches |S11| < -20 dB.\n');
end

%% Plotting
figure('Name', 'S11 Frequency Behavior', 'Color', 'w');

plot(f_norm, S11_1_dB, 'b', 'LineWidth', 1.5); hold on;
plot(f_norm, S11_2_dB, 'r--', 'LineWidth', 1.5);
yline(-20, 'k:', 'LineWidth', 1.5, 'Label', '-20 dB target');

% Formatting the plot
ylim([-50 0]);
xlim([0 2]);
grid on;
title('Frequency Behavior of QWT');
xlabel('Normalized Frequency (f/f_0)');
ylabel('|S_{11}| (dB)');
legend('Case 1: Z_L = 100 \Omega', 'Case 2: Z_L = 200 \Omega',
   'Location', 'south');
```

A.2 Problem nr.2 MATLAB code

```
clear; clc; close all;

%% Define Parameters
Z0 = 50;            % Reference/Source impedance
ZL = 100;           % Load impedance
Zc1 = 60;           % Characteristic impedance of line 1
Zc2 = 84;           % Characteristic impedance of line 2

f_norm = linspace(0, 2, 10000); % Normalized frequency f/f0
theta = (pi / 2) * f_norm;      % Electrical length
```

```matlab

%% Calculate S11 using ABCD Matrices
% ABCD matrix elements for Line 1
A1 = cos(theta);
B1 = 1j * Zc1 * sin(theta);
C1 = 1j * sin(theta) / Zc1;
D1 = cos(theta);

% ABCD matrix elements for Line 2
A2 = cos(theta);
B2 = 1j * Zc2 * sin(theta);
C2 = 1j * sin(theta) / Zc2;
D2 = cos(theta);

% In order to cascade the two sections we have to multiply
% the 2 matrixes: [ABCD_total] = [ABCD_1] * [ABCD_2]
A_tot = A1 .* A2 + B1 .* C2;
B_tot = A1 .* B2 + B1 .* D2;
C_tot = C1 .* A2 + D1 .* C2;
D_tot = C1 .* B2 + D1 .* D2;

% Calculate Input Impedance and S11
Zin_p2 = (A_tot .* ZL + B_tot) ./ (C_tot .* ZL + D_tot);
S11_p2 = (Zin_p2 - Z0) ./ (Zin_p2 + Z0);
S11_p2_dB = 20 * log10(abs(S11_p2));

%% Determine Bandwidth (S11 < -20 dB)
idx_p2 = find(S11_p2_dB < -20);
if ~isempty(idx_p2)
    f1_p2 = f_norm(idx_p2(1));
    f2_p2 = f_norm(idx_p2(end));
    BW_p2 = f2_p2 - f1_p2;
    fprintf('Problem 2 (Two-Section):\n');
    fprintf('  Interval [f1, f2] / f0: [%.3f, %.3f]\n', f1_p2,
       f2_p2);
    fprintf('  Fractional Bandwidth: %.2f%%\n\n', BW_p2 * 100);
end

%% Plotting
```

```
figure('Name', 'Task1', 'Color', 'w');
plot(f_norm, S11_p2_dB, 'r-', 'LineWidth', 1.5);
hold on;
yline(-20, 'k:', 'LineWidth', 1.5, 'Label', '-20 dB target');

% Formatting the plot
ylim([-50 0]);
xlim([0 2]);
grid on;
title('Frequency Behavior of Matching Circuits (Z_L = 100 \
   Omega)');
xlabel('Normalized Frequency (f/f_0)');
ylabel('|S_{11}| (dB)');
hold off;
```